

Implicit ABBA Jacobian formulas

Implicit-function and stage-increment factorizations

1 Notation

Let

$$s := \frac{h}{2}, \quad W_i := D_z f(\text{ABBA stage point}_i, \text{stage time}_i), \quad S := W_2 + W_3.$$

Every W_i is a 2×2 vector-field Jacobian. Matrix-product order must be preserved because the W_i do not commute in general.

The implicit multiplier is a function of the physical input,

$$\mu_n = \mu_h(z_n),$$

and its total derivative is denoted by

$$M := D\mu_h(z_n).$$

The two displaced initial copies therefore satisfy

$$Du^{(0)} = I + M, \quad Dv^{(0)} = I - M.$$

2 Residual derivatives K and L

For the projection residual $R(z, \mu)$, define

$$K := D_\mu R, \quad L := D_z R.$$

In terms of the four stage Jacobians,

$$K = 4I - s(W_1 + W_2 + W_3 + W_4) + s^2(W_4 S + S W_1) - s^3 W_4 S W_1,$$

and

$$L = s(W_1 - W_2 - W_3 + W_4) + s^2(W_4 S - S W_1) + s^3 W_4 S W_1.$$

Equivalently, after replacing $S = W_2 + W_3$,

$$\begin{aligned} K &= 4I - s(W_1 + W_2 + W_3 + W_4) \\ &\quad + s^2[W_4(W_2 + W_3) + (W_2 + W_3)W_1] \\ &\quad - s^3 W_4(W_2 + W_3)W_1, \\ L &= s(W_1 - W_2 - W_3 + W_4) \\ &\quad + s^2[W_4(W_2 + W_3) - (W_2 + W_3)W_1] \\ &\quad + s^3 W_4(W_2 + W_3)W_1. \end{aligned}$$

Differentiating the exact-root identity

$$R(z, \mu_h(z)) = 0$$

gives

$$L + KM = 0.$$

If K is invertible, then

$$M = -K^{-1}L.$$

In an implementation this inverse is not formed explicitly: the matrix system $KM = -L$ is solved instead.

3 Total derivatives of the four increments

The physical mean can be written as

$$z_{n+1} = z_n + \frac{h}{4}(k_1 + k_2 + k_3 + k_4).$$

The total stage derivatives, including the dependence of μ_h on z_n , obey

$$\begin{aligned} Dk_1 &= W_1(I - M), \\ Du^{(1)} &= I + M + sDk_1, \\ Dk_2 &= W_2Du^{(1)}, \\ Dk_3 &= W_3Du^{(1)}, \\ Dv^{(2)} &= I - M + s(Dk_2 + Dk_3), \\ Dk_4 &= W_4Dv^{(2)}. \end{aligned}$$

Eliminating the intermediate derivatives gives the four requested expressions:

$$Dk_1 = W_1(I - M),$$

$$Dk_2 = W_2[I + M + sW_1(I - M)],$$

$$Dk_3 = W_3[I + M + sW_1(I - M)],$$

and

$$Dk_4 = W_4\left\{I - M + s(W_2 + W_3)[I + M + sW_1(I - M)]\right\}.$$

Consequently, the stage-increment factorization is

$$J_n^{\text{inc}} = I + \frac{h}{4}(Dk_1 + Dk_2 + Dk_3 + Dk_4).$$

4 Equality with the implicit-function factorization

For completeness, define

$$P = I + s(W_1 + W_4) + s^2W_4S + s^3W_4SW_1,$$

$$Q = 2I - s(W_1 + W_4) + s^2W_4S - s^3W_4SW_1.$$

The derivative of the corrected first copy is

$$J_n^{\text{copy}} = P + QM.$$

Direct symbolic expansion of the stage-increment recurrence gives the identity

$$J_n^{\text{inc}} - (P + QM) = -\frac{1}{2}(L + KM).$$

At the implicit root, $L + KM = 0$, and hence

$$J_n^{\text{inc}} = P + QM.$$

Finally, substituting $M = -K^{-1}L$ yields

$$J_n^{\text{inc}} = I + \frac{h}{4} \sum_{i=1}^4 Dk_i = P - QK^{-1}L = J_n^{\text{IFT}}.$$

Thus the two calculations are symbolically different derivations and computational factorizations of the same ideal-root physical Jacobian.